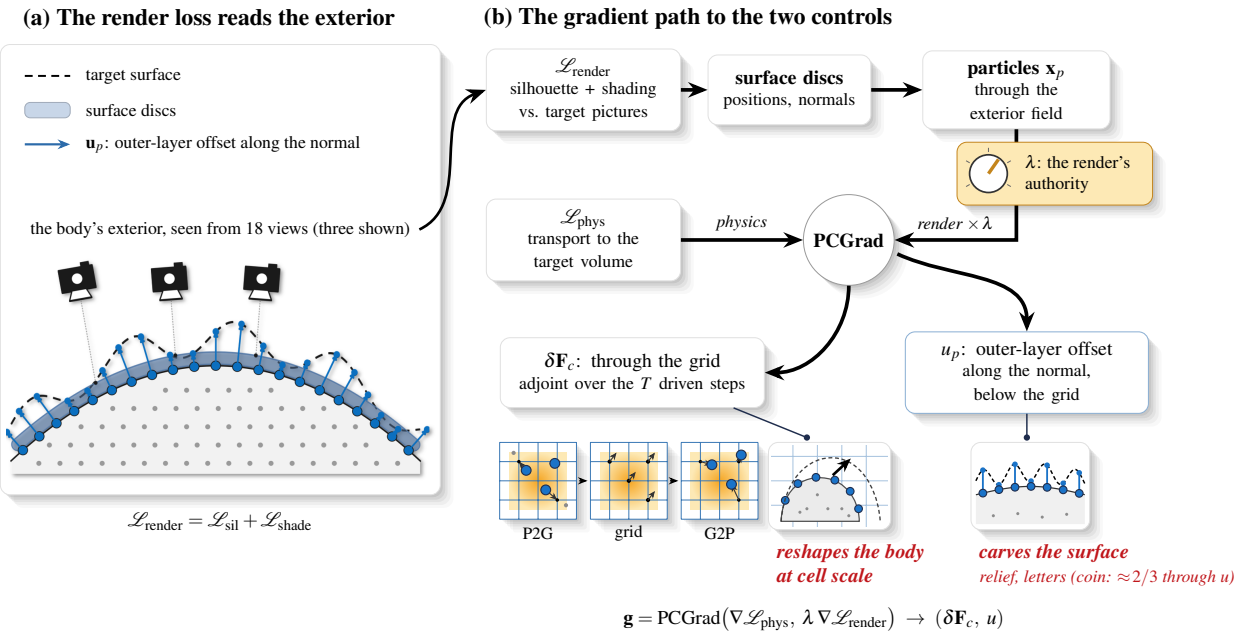

(a) The render loss reads the exterior
target surface
surface discs
$\mathbf{u}_p$: outer-layer offset along the normal
the body's exterior, seen from 18 views (three shown)
$\mathscr{L}_{\text{render}} = \mathscr{L}_{\text{sil}} + \mathscr{L}_{\text{shade}}$
(b) The gradient path to the two controls
$\mathscr{L}_{\text{render}}$ silhouette + shading vs. target pictures
surface discs positions, normals
particles $\mathbf{x}_p$ through the exterior field
$\lambda$: the render's authority
$\mathscr{L}_{\text{phys}}$ transport to the target volume
physics
PCGrad
render $\times\,\lambda$
$\delta\mathbf{F}_c$: through the grid adjoint over the $T$ driven steps
$u_p$: outer-layer offset along the normal, below the grid
P2G
grid
G2P
reshapes the body at cell scale
carves the surface
relief, letters (coin: $\approx 2/3$ through $u$)
$\mathbf{g} = \text{PCGrad}\big(\nabla\mathscr{L}_{\text{phys}},\ \lambda\,\nabla\mathscr{L}_{\text{render}}\big) \ \rightarrow\ (\delta\mathbf{F}_c,\ u)$